

FLOWERING PLANTS

With over 300,000 species, flowering plants – or angiosperms – are by far the largest group of plants, as well as the most diverse. They play a vital role in most land-based ecosystems, producing food and shelter for animals and many other living things.

DIVISION	ANGIOSPERMAE
CLADES	10
ORDERS	64
FAMILIES	416
SPECIES	About 304,000



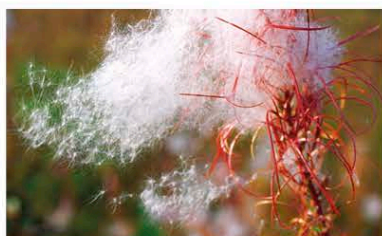
Wind-pollinated flowers such as hazel catkins shed clouds of pollen grains into the air. They are rarely brightly coloured.



Animal-pollinated flowers are usually conspicuous, and have sticky pollen grains. Hummingbirds transfer pollen as they feed.



Fleshy fruits evolved to attract animals. Wild cucumbers are eaten by antelopes, which scatter the seeds in their droppings.



Dry fruits often burst open when their seeds are ripe. This willowherb is scattering its fluffy seeds into the wind.

The first flowering plants evolved 140 million years ago, which makes them relative latecomers to life on Earth. Since then, they have become the dominant forms of plant life. The smallest flowering plants are not much bigger than a pinhead, but this group also includes all broadleaved trees, and a variety of other species, from cacti and grasses to orchids and palms.

Flowering plants share several key features that help to make them successful. Foremost among these are flowers – actually a collection of highly modified leaves. In most flowers, the outermost layers are the sepals, then the petals. These surround the male stamens, which make pollen, and the female carpels, which collect pollen arriving from similar flowers, so that their female ovules can develop into seeds. Some flowers spread their pollen through the air, but many more use animals to transfer pollen for them. In these showy species, flowers lure animals to sip the sugary nectar in the innermost part of the flower, brushing past the pollen on the way. These plants mostly attract insects, but birds and bats are also important pollinators.

STRATEGIES FOR SPREADING

Flowering plants are not unique in making seeds, but they are the only plants that produce them in fruits. A fruit develops from a flower's ovaries – the closed chambers that house the seeds, close to the flower's central stem. Fruits have a double function: to protect the seeds and to help them spread. Fleshy fruits do this by attracting animals, which feed and scatter the seeds in their excrement. Dry fruits work in a variety of ways. Some burst open when their seeds are ripe. Others have hooks that latch onto clothing, skin, or fur. Still more drift in water or through the air. Many flowering plants spread by the way they grow, often producing new plants from creeping stems. This can create vast, interconnected clones. The largest examples, formed by North American aspen trees, stretch over 40 ha (100 acres), and may be 10,000 years old.

ATTRACTING ATTENTION >

Hellebores use colourful sepals, rather than petals, to attract pollinating insects – or multiple sepals in this garden variety.

